

EXERCISES: Will Free

```
In [4]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [3]: def trap(x_data, y_data):
    if type(x_data) == pd.core.series.Series:
        x_data = np.asarray(x_data) # asarray is converting pandas into an array
        y_data = np.asarray(y_data)
    dx = (x_data[-1] - x_data[0]) / len(x_data)
    s = dx / 2 * ((y_data[0]) + (y_data[-1]))
    s = s + np.sum(y_data[1:-2] * dx)
    return(s)

def simpsons(func, a, b, steps):

    if steps % 2 != 0:
        steps = steps + 1
        print('I added one step')

    h = (b - a) / steps
    x = np.linspace(a, b, steps + 1)
    y = func(x)
    s = h / 3 * (func(a) + func(b))
    s = s + (h / 3) * (4 * np.sum(y[1:steps:2]) + 2 * np.sum(y[2:steps-1:2]))
    return(s)
```

5.1:

```
In [5]: df = pd.read_csv('../data/velocities.txt', sep='\t', names=['t', 'v'])
df
```

```
Out[5]:
```

	t	v
0	0	0.000000
1	1	0.069478
2	2	0.137694
3	3	0.204332
4	4	0.269083
...	...	...
96	96	0.223073
97	97	0.254244
98	98	0.283753
99	99	0.311479
100	100	0.337308

101 rows × 2 columns

```
In [7]: integrals = [0]

for i in range(1,101):
    w = trap(df['t'][0:i], df['v'][0:i])
    integrals.append(w)
```

```
In [8]: df['t'][0:5]
```

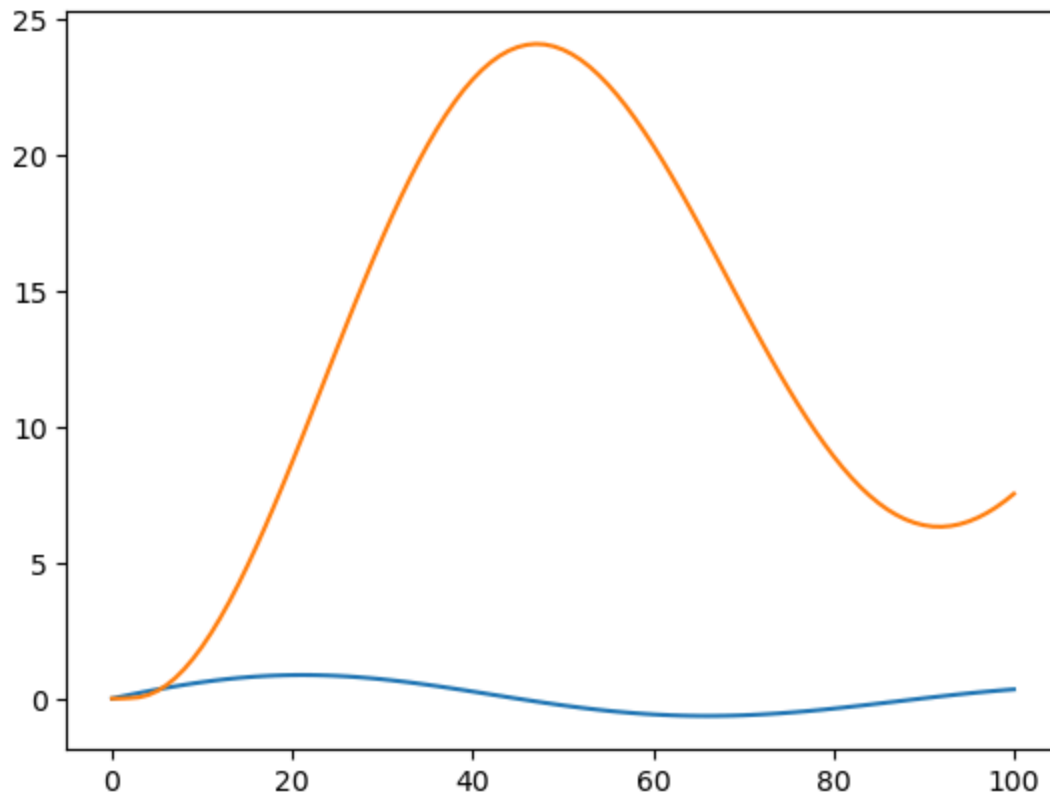
```
Out[8]: 0    0
        1    1
        2    2
        3    3
        4    4
        Name: t, dtype: int64
```

```
In [9]: df['t'].index
```

```
Out[9]: RangeIndex(start=0, stop=101, step=1)
```

```
In [10]: fig, ax = plt.subplots()
ax.plot(df['t'],df['v'])
ax.plot(df['t'], integrals)
```

```
Out[10]: [<matplotlib.lines.Line2D at 0x140c97d90>]
```



5.2

```
In [5]: def simpsons(func, a, b, steps):
        if steps%2!=0:
            return('that doesnt work -- use an even number of steps')

        h = (b-a)/steps
        x = np.linspace(a, b, steps)
        y = func(x)
        s = h/3*(func(a) +func(b))# (s stands for sum)
        s = s + 4 * np.sum(y[1:steps-1:2]*h/3) + 2*np.sum(y[2:steps-2:2]*h/3)# r
        return(s)
```

```
In [6]: def f(x):
        return(x**4-2*x+1)
```

```
In [7]: simpsons(f, 0, 2, 10000000)
```

```
Out[7]: 4.3999969600009985
```

5.3

```
In [22]: def simpsons(func, a, b, steps):
        if steps%2!=0:
            return('that doesnt work -- use an even number of steps')
```

```

h = (b-a)/steps
x = np.linspace(a, b, steps+1)
y = np.array([f(xi) for xi in x])
s = h/3*(y[0] + y[-1] + 4*sum(y[1:-1:2]) + 2*sum(y[2:-2:2]))+func(a) +fur

return(s)

```

```

In [23]: def f(t):
          return(np.exp(-t**2))

```

```

In [36]: x=1
          result = simpsons(f, 0, x, 10)
          print(f"integral of e**{-t**2} from 0 to x = " + str(result))
          print(f"exact value: {x**3/3}")

```

integral of e**{-t**2} from 0 to x = 0.7924209296268249
exact value: 0.3333333333333333

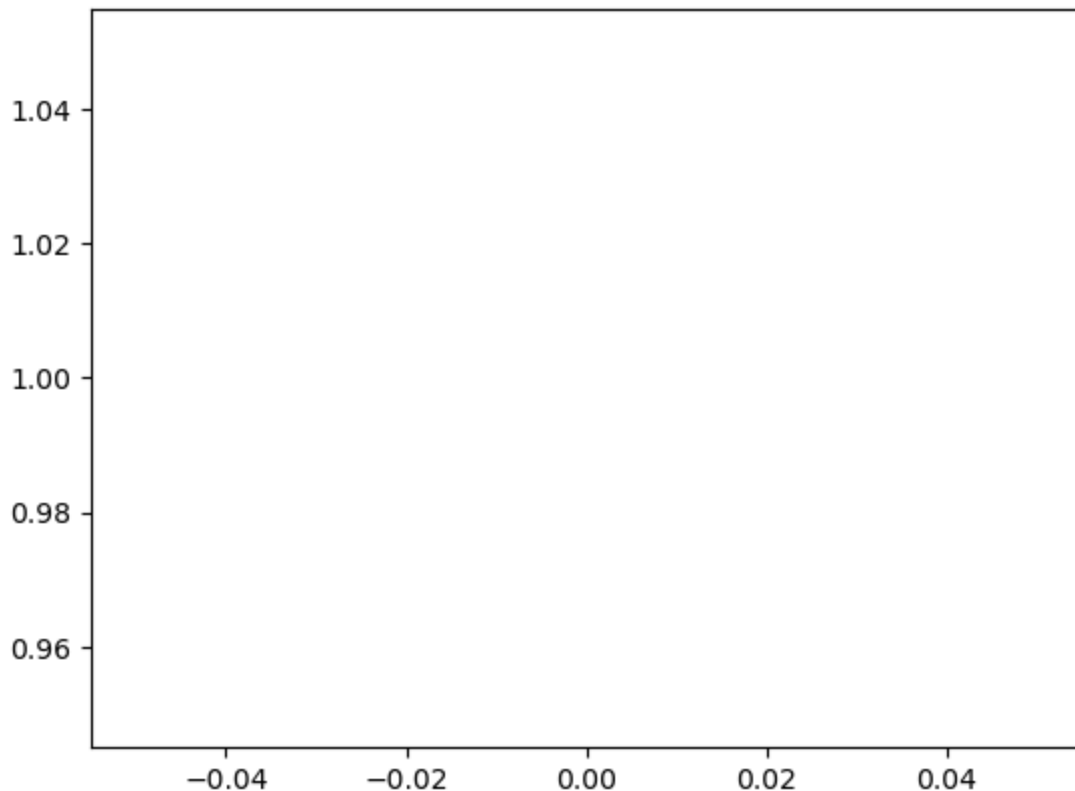
```

In [25]: fig0 = plt.figure()
          ax0 = fig0.add_subplot()

          ax0.plot(x)

```

Out[25]: [<matplotlib.lines.Line2D at 0x7e70a4277b00>]



In []:

In []:

```
In [21]: # Everything worked but nothing graphed???
```

3.8

```
In [33]: data = pd.read_csv('../data/millikan.txt', sep=' ', names=['x','y'])
```

```
In [34]: data
```

```
Out[34]:
```

	x	y
0	5.487400e+14	0.53090
1	6.931000e+14	1.08420
2	7.430700e+14	1.27340
3	8.219300e+14	1.65980
4	9.607400e+14	2.19856
5	1.184000e+15	3.10891

```
In [35]: data = np.loadtxt('../data/millikan.txt')
data = data.T
x = data[0]
y = data[1]
N = len(x)

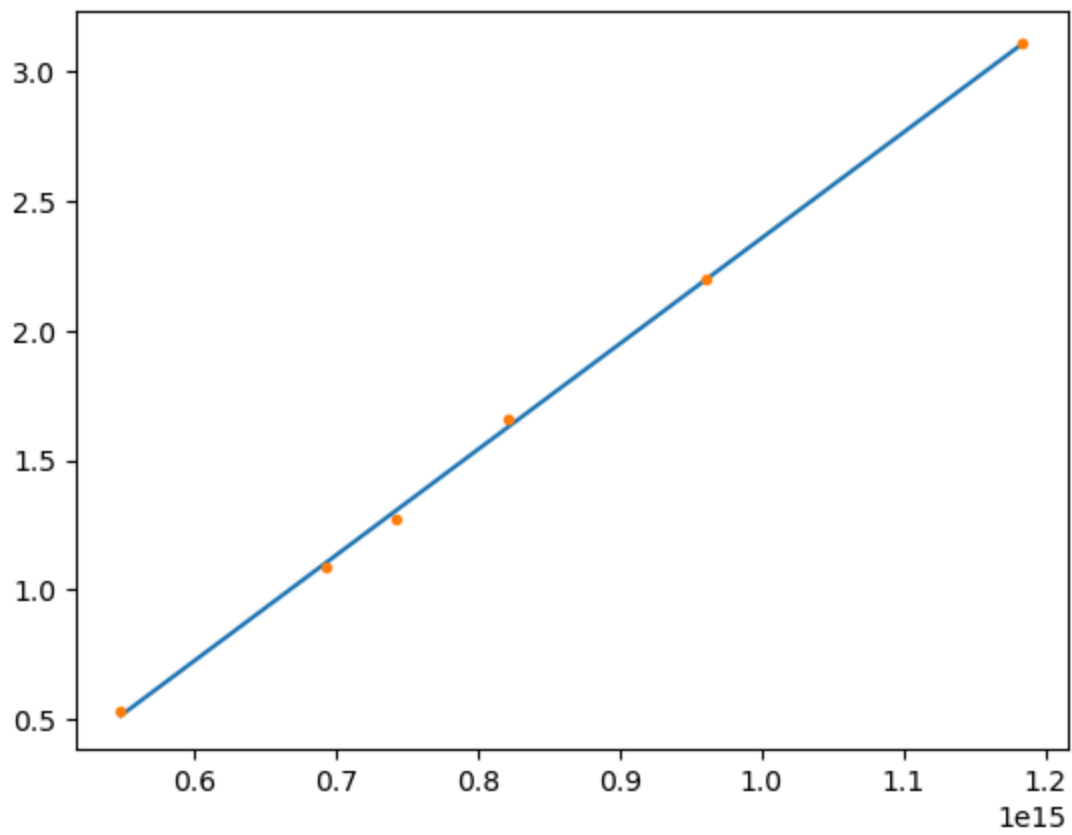
Ex = np.sum(x)/N
Ey = np.sum(y)/N
Exx = np.sum(x**2)/N
Exy = np.sum(x*y)/N
m = (Exy - (Ex*Ey)) / ((Exx - Ex**2))
c = ((Exx*Ey) - (Ex*Exy)) / (Exx - Ex**2)

fy = m*x+c
plt.plot(x, fy)
plt.plot(x, y, '.')
```

print(m,c)

print(m*1.602e-19)

```
4.088227358517516e-15 -1.7312358039813558
6.549340228345061e-34
```



In []: